Week 6 Lab: Final Controller Validation and Project Preparation

ENGR 3303 – Controls Engineering
Fall 2026

> **Main goal.** Validate the final motor-speed controller using several reference speeds and repeated trials, create final Python figures, and prepare a reliable Week 7 demonstration.

Required Python Packages

```
pip install pyserial pandas numpy matplotlib scipy
```

Course encoder calibration: Use `COUNTS_PER_OUTPUT_REV = 494.0` for Encoder A counted with `RISING` edges only. Recalibrate only if the motor or encoder-counting method changes.

Part 0 – Frequency-Response Visualization

Before hardware testing, run:

`week6_frequency_response.py`

Enter your Week 2 values:

$$K = _____, \qquad \tau = _____ \text{ s.}$$

The script produces:

- magnitude vs. frequency;
- phase vs. frequency;
- calculated corner frequency.

Save the plots if you want to use them in the final presentation. This is a conceptual visualization only; no hand Bode plot is required.

Lab 6A – Final Performance Validation

Suggested time: 75 minutes

Part 1 – Record Final Controller Settings

$$\boxed{K_p = _____, \quad K_i = _____, \quad T_s = _____ \text{ s}, \quad \alpha = _____}$$

Integral clamp:

$$\boxed{I_{\min} = _____, \qquad I_{\max} = _____}$$

Part 2 – Choose Three Reference Speeds

Choose instructor-approved values that are safe and achievable.

$$r_1 = _____ \text{ RPM}$$

$$r_2 = _____ \text{ RPM}$$

$$r_3 = _____ \text{ RPM}$$

Possible example values are 80, 120, and 150 RPM, but use values appropriate for your hardware.

Part 3 – Upload Final Controller Code

Open:

`week6_final_controller.ino`

For each test, change only:

`REFERENCE_RPM`.

Keep the final controller settings fixed.

Part 4 – Record Each Test

Run:

`week6_record_final_test.py`

Save files with clear names such as:

`final_ref80.csv`

`final_ref120.csv`

`final_ref150.csv`.

Part 5 – Compare Reference Speeds

Run:

`week6_compare_references.py`

Complete:

Reference	Final RPM	Error	Overshoot	Max PWM	Observation

Lab 6A checkoff: Show the instructor the three-reference comparison plot and your completed performance table.

Lab 6B – Repeatability and Project Preparation

Suggested time: 75 minutes

Part 1 – Repeat the Main Test

Choose the reference speed you plan to use for the Week 7 demo:

$$\boxed{r_{\text{demo}} = _____ \text{ RPM}}$$

Run the same test three times.
Save:

`demo_trial1.csv`

`demo_trial2.csv`

`demo_trial3.csv`.

Use the comparison script from Week 5 or adapt the supplied Week 6 scripts.

Part 2 – Optional Safe Disturbance Test

With instructor approval only, apply a small safe disturbance.
Record:

- approximate time of disturbance;
- minimum RPM after disturbance;

- maximum PWM after disturbance;
- whether the speed recovered near the reference.

> Do not touch the spinning wheel directly. The instructor should approve the disturbance method before the test.

Part 3 – Create Final Project Figures

Run:

`week6_make_project_figures.py`

The script is designed to create up to three clean figures:

1. P-control response;
2. PI-control response;
3. final filtered-controller response.

Update the filenames at the top of the script to match your saved CSV files.

Part 4 – Select Presentation Figures

Choose only the plots that support your story.
Recommended:

- Week 2 model comparison;
- P vs. PI comparison;
- final controller tracking;
- optional repeated-trial or disturbance figure.

Part 5 – Final Demo Checklist

Mark each item when ready.

☐ Wiring is correct and secure.

☐ Final Arduino code is saved.

☐ Final K_p, K_i, T_s, and α are recorded.

☐ Demo reference speed is recorded.

☐ Python recorder works.

☐ Backup CSV data are saved.

☐ Final plots are saved.

☐ Team members know their demo roles.

Questions to Submit

1. What final K_p, K_i, T_s, and α did your team use?
2. What three reference speeds did you test?
3. At which reference speed was tracking most accurate?
4. At which reference speed was PWM saturation most significant?
5. Did controller performance change with reference speed? Explain.
6. How repeatable were the three demo trials?
7. What is the corner frequency of your Week 2 first-order motor model?
8. What does the frequency-response magnitude plot say about fast motor commands?
9. What plot will be most important in your final presentation and why?
10. What is the main limitation of your current controller or hardware?

What to Submit

One submission per team:

- three-reference CSV data;
- repeated demo-trial CSV data;
- three-reference comparison plot;
- final controller performance table;
- selected final project figures;
- answers to Questions 1–10;
- final Arduino and Python files.